Natural Language Processing
Lab 5: Classification with Linear Models

Lonny Chen: l.chen@hertie-school.org
GRAD-E1282, Fall 2026
30 September - 1 October 2026

Today’s Agenda

1. Pipeline Choices
    a. Preprocessing
    b. Representation
    c. Modeling: Linear (Basic)
2. Performance metrics
    a. Dataset Examples
    b. Cost Considerations

Pipeline Choices

Preprocessing

Representation

Modeling

Pipeline Choices: Preprocessing

Preprocessing | Representation | Modeling

Tokenization:

- Character
- Word
- Subword

Normalization:

- Case folding
- Alphabetical / numeric / punctuation
- Stopwords
- Lemmatizaton

Pipeline Choices: Representation

Preprocessing

Representation

Modeling

Sparse counts/weights:

- Bag of Words
- Bag of N-grams
- TF-IDF

Dense embeddings:

- Word2Vec
- Doc2Vec
- …

Pipeline Choices: Representation

Preprocessing

Representation

Modeling

Sparse counts/weights:

- Interpretable features
- Keyword-driven tasks
- Good for large labeled N
- No learned semantics

Dense embeddings:

- Word2Vec
- Doc2Vec
- …

Pipeline Choices: Representation

Preprocessing

Representation

Modeling

Sparse counts/weights:

- Interpretable features
- Keyword-driven tasks
- Good for large labeled N
- No learned semantics

Dense embeddings:

- Semantic-driven tasks learned from corpus
- Can help with small N
- Unintrepretable features

Pipeline Choices: Modeling

Preprocessing

Representation

Modeling

Naive Bayes:

- Generative (re-construct)
- Class-feature probabiliities

Logistic Regression:

- Discriminative (divide)
- Log-odds of class probability vs. linear X

Support Vectors:

- Discriminative (divide)
- Linear class boundary based on support vectors of boundary with margin

Pipeline Choices: Modeling

Preprocessing	Representation	Modeling

Naive Bayes:

- Generative (re-construct)
- Class-feature probabiliities
- Fast to run
- Inspect word log-prob
- Less effective with embeddings
- ”Naive” assumption of feature independence
- Linear only

Logistic Regression:

- Discriminative (divide)
- Log-odds of class probability vs. linear X

Support Vectors:

- Discriminative (divide)
- Linear class boundary based on support vectors of boundary with margin

Pipeline Choices: Modeling

Preprocessing

Representation

Modeling

Naive Bayes:

- Generative (re-construct)
- Class-feature probabiliities
- Fast to run
- Inspect word log-prob
- Less effective with embeddings
- ”Naive” assumption of feature independence
- Linear only

Logistic Regression:

- Discriminative (divide)
- Log-odds of class probability vs. linear X
- Provides certainty of fit
- Can calibrate probability threshold → class
- Inspect word coefficients
- Linear only

Support Vectors:

- Discriminative (divide)
- Linear class boundary based on support vectors of boundary with margin

Pipeline Choices: Modeling

Preprocessing

Representation

Modeling

Naive Bayes:

- Generative (re-construct)
- Class-feature probabiliities
- Fast to run
- Inspect word log-prob
- Less effective with embeddings
- ”Naive” assumption of feature independence
- Linear only

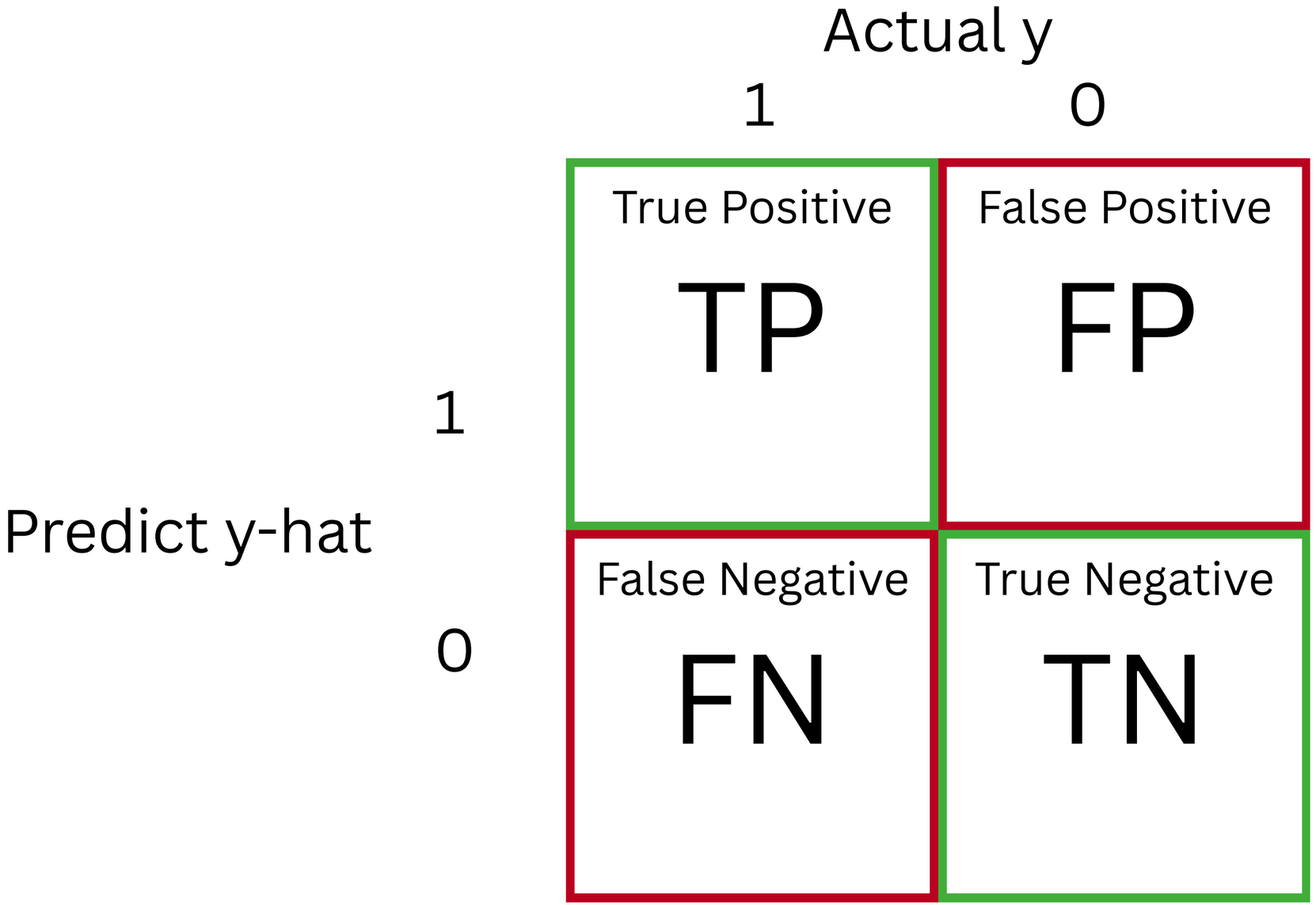
Logistic Regression:

- Discriminative (divide)
- Log-odds of class probability vs. linear X
- Provides certainty of fit
- Can calibrate probability threshold → class
- Inspect word coefficients
- Linear only

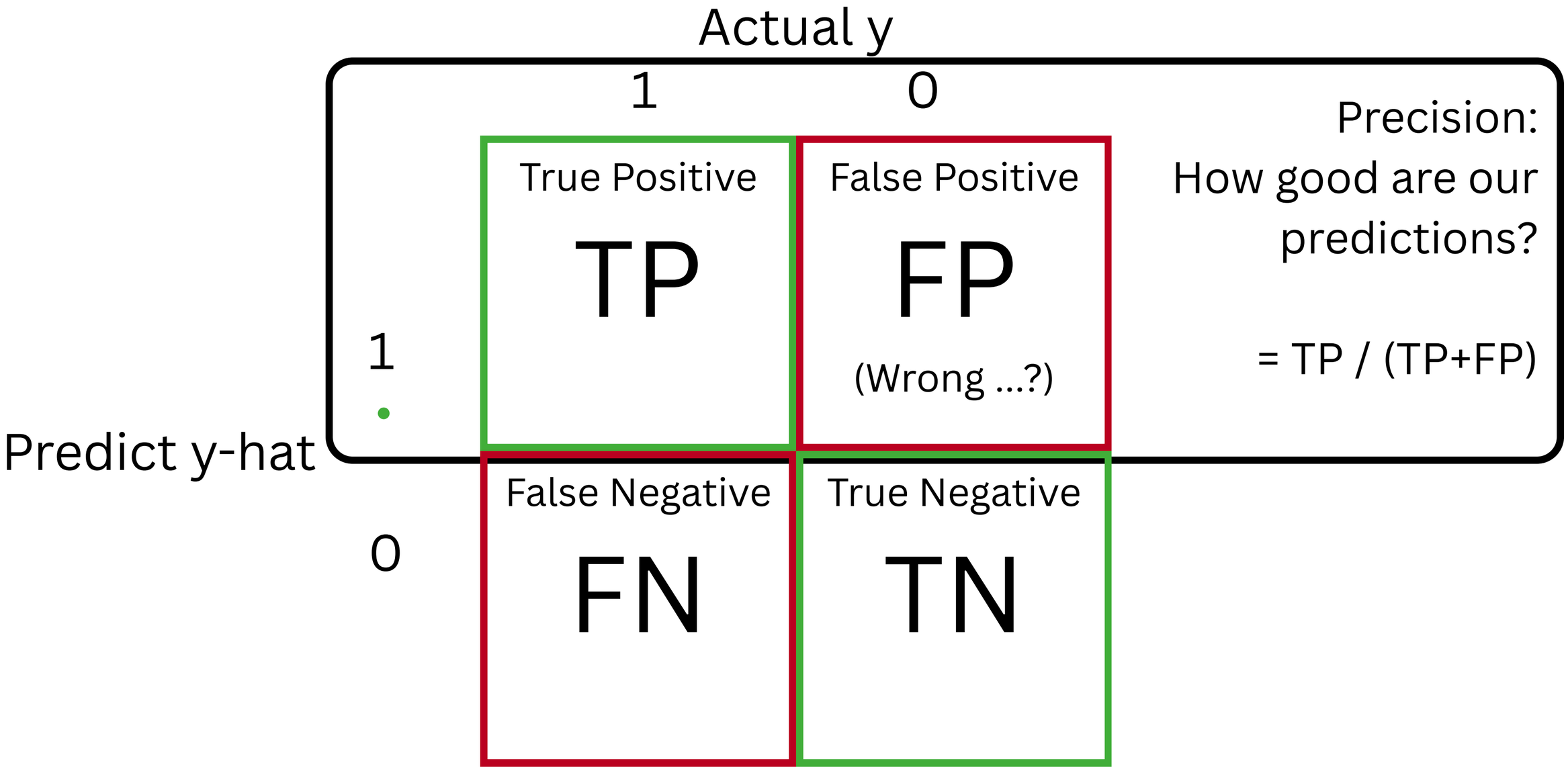
Support Vectors:

- Discriminative (divide)
- Linear class boundary based on support vectors of boundary with margin
- Good with many features (focuses on boundary)
- inspect word coefficients
- Also non-linear with kernels
- Slow to train/tune (large N)

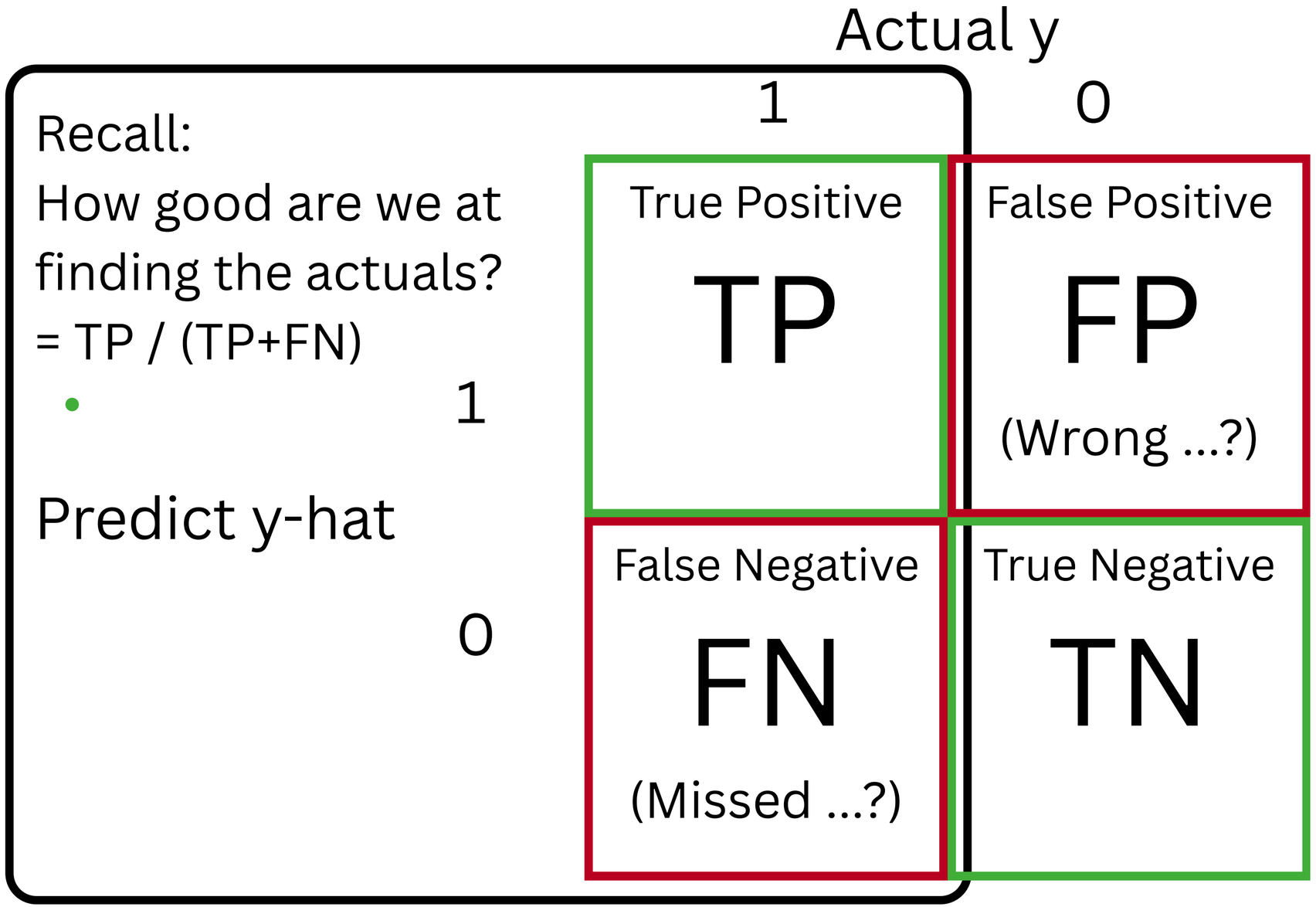
Performance Metrics



Performance Metrics



Performance Metrics



Example 1: Infectious Virus Treatment

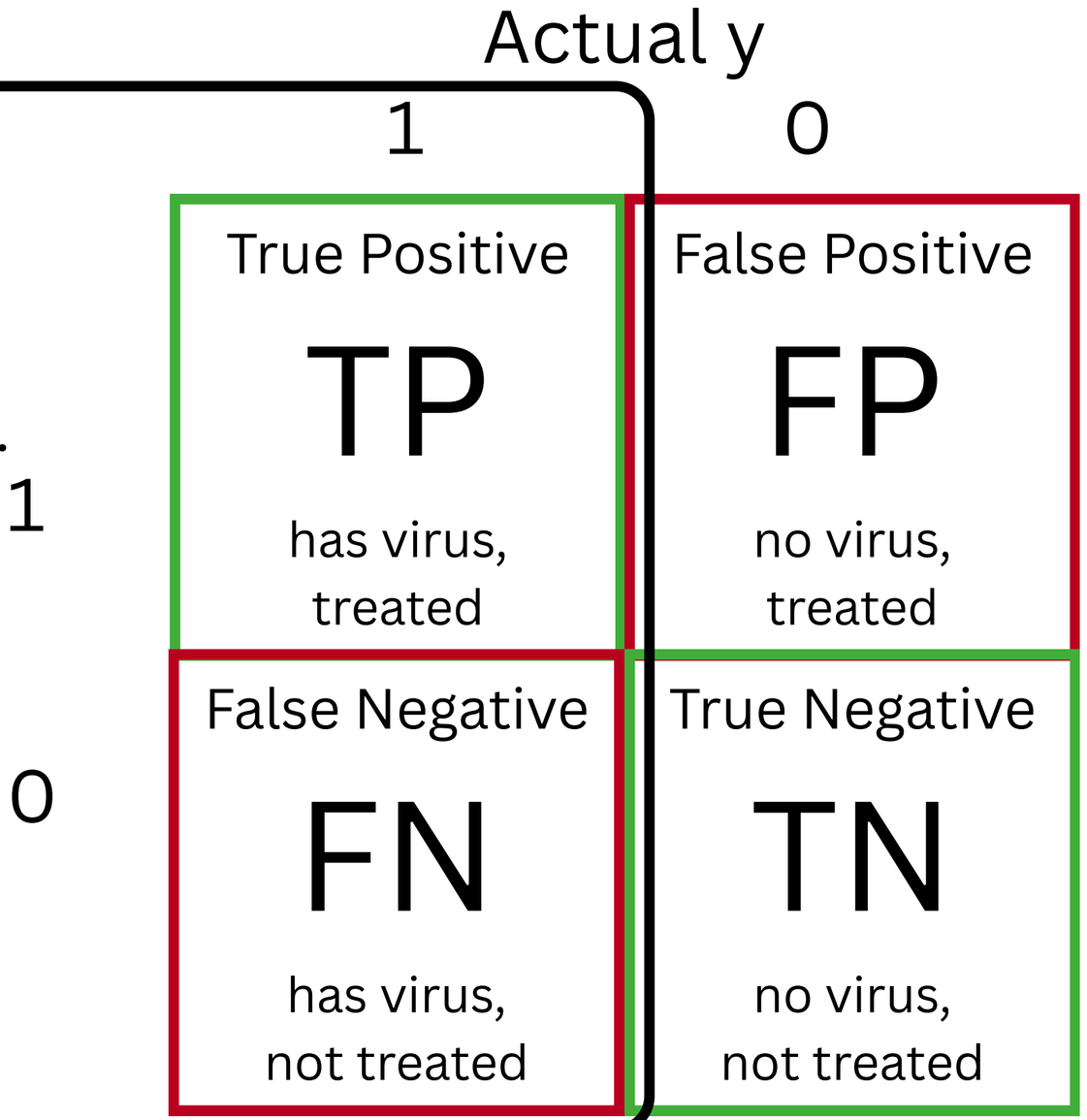
		Actual y = 1	Actual y = 0
Predict y-hat	1	True Positive TP has virus, treated	False Positive FP no virus, treated
	0	False Negative FN has virus, not treated	True Negative TN no virus, not treated

Example 1: Infectious Virus Treatment

High Recall:

Avoid FN: missed treatment of people with virus.

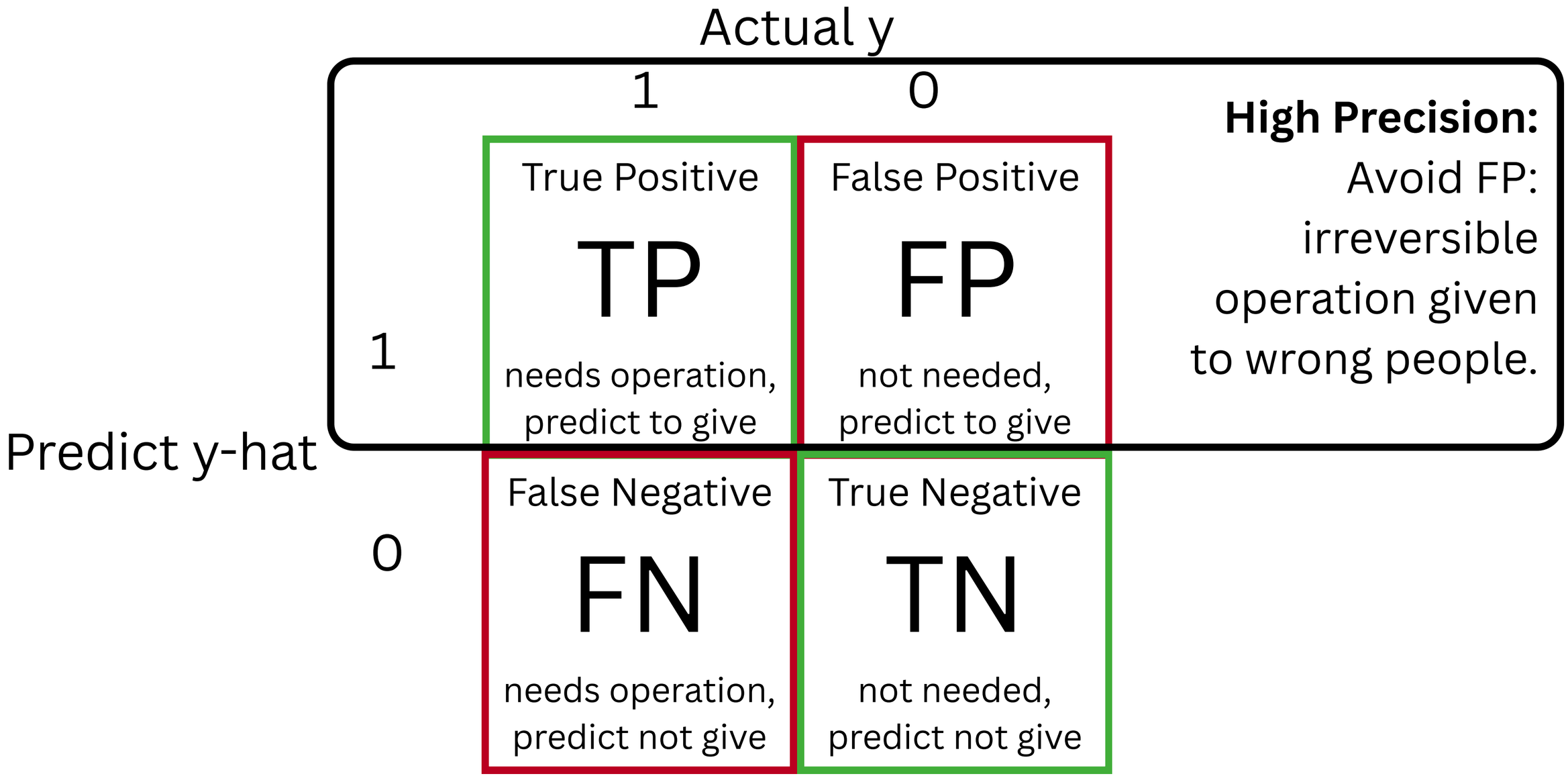
- Predict y-hat 1
- Predict y-hat 0



Example 2: Irreverisble Operation

Predict y-hat	Actual y = 1	Actual y = 0
1	True Positive **TP** needs operation, predict to give	False Positive **FP** not needed, predict to give
0	False Negative **FN** needs operation, predict not give	True Negative **TN** not needed, predict not give

Example 2: Irreverisble Operation



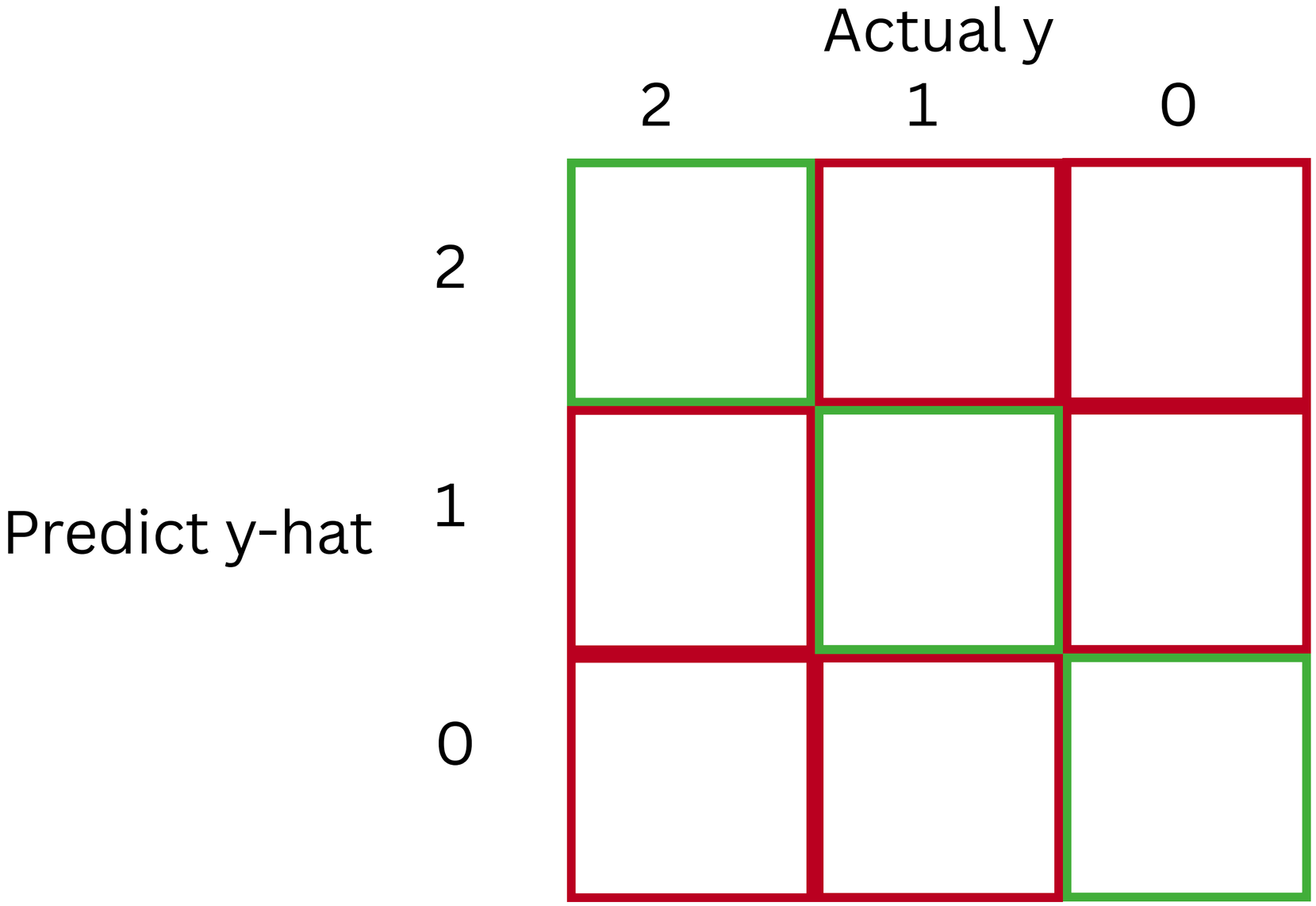
Dataset 1: Spam Detection

		Actual y = 1	Actual y = 0
Predict y-hat	1	True Positive TP spam, marked	False Positive FP OK email, marked spam
	0	False Negative FN spam, not marked	True Negative TN OK email, not marked

Dataset 2: Machine-Gen. Text Detection

Predict y-hat \ Actual y	1	0
1	True Positive TP Machine, flagged machine	False Positive FP Human, flagged machine
0	False Negative FN Machine, not flagged	True Negative TN Human, not flagged

Dataset 3: Sentiment Analysis



Next Week: Topic Modeling!

